TITLE PAGE

- **Problem Statement ID –** Question 4
- **Problem Statement Title-** AI-Powered Behavioral Anomaly Detection for Cybersecurity
- **Theme-** Cybersecurity
- **PS Category-** Software
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SOC Sentinel: Dual-Model Behavioral Anomaly Engine

Proposed Solution:-

- Detailed explanation of the proposed solution:-
  - A localized, real-time threat detection engine designed to analyze access-log telemetry without relying on static signatures.
  - Utilizes a two-stage machine learning ensemble: an unsupervised model establishes a behavioral baseline, and a supervised classifier categorizes specific threat vectors.
  - Translates mathematical model confidence directly into an actionable 0-100 Risk Score for Security Operations Center (SOC) analysts.
- How it addresses the problem:-
  - **Imbalance:** Bypasses the class imbalance problem by using outlier detection (Isolation Forest) to isolate anomalies before classification.
  - **UX/Explainability:** Replaces binary "alert/no-alert" outputs with context-rich explanations (e.g., AI confidence percentages, specific attack labels) inside a filterable React dashboard
- Innovation and uniqueness of the solution:-
  - **Hardware-Efficient:** Explicitly designed to run inference on edge-level hardware (tested on an Intel Core i3) without heavy recurrent networks.
  - **Cold-Start Resilient:** Evaluates events against global telemetry baselines rather than requiring historical data for every new user or device.

TECHNICAL APPROACH

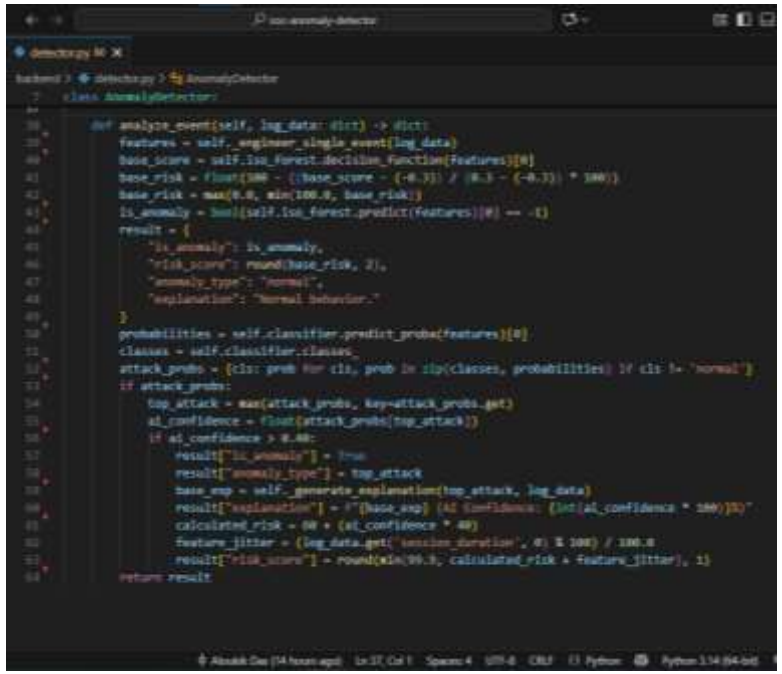
- Technologies to be used:-
  - **Core ML:** Python, Scikit-Learn (Isolation Forest, Random Forest Classifier), Pandas, NumPy.
  - **Backend & Data:** FastAPI (async log streaming), SQLite, SQLAlchemy.
  - **Frontend UI:** React.js, Tailwind CSS, Recharts (dynamic telemetry visualization), Lucide Icons.
- Methodology and process for implementation:-
  - **Phase 1 (Data Ingestion):** Pandas-driven generator creates synthetic user/device profiles and injects exact threat taxonomies (Brute Force, Impossible Travel, Lateral Movement).
  - **Phase 2 (Feature Engineering):** Raw logs converted into behavioral vectors (session duration, geo-velocity, failure rates).
  - **Phase 3 (Inference Pipeline):**
    - ***Step A:*** Isolation Forest calculates statistical deviation from normal traffic.
    - ***Step B:*** If deviation threshold is met, Random Forest classifies the exact attack type.
  - **Phase 4 (Actionable Output):** FastAPI pushes scored alerts to the React UI, rendering real-time radar animations and generating downloadable CSV threat reports.

FEASIBILITY AND VIABILITY

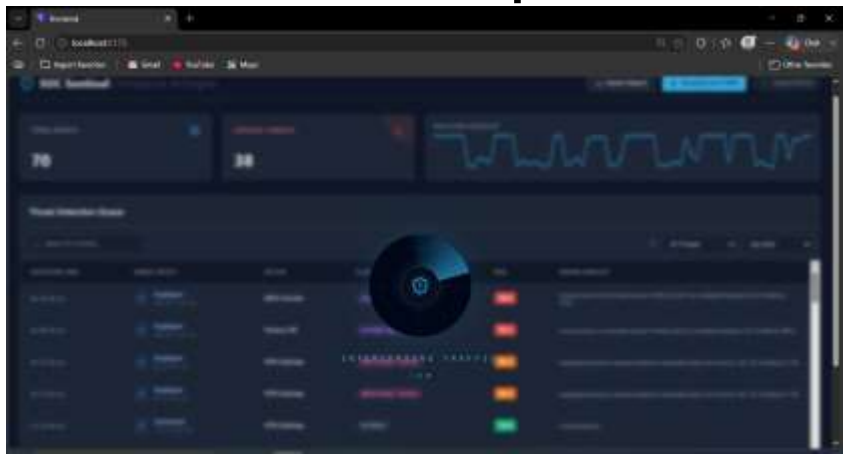
- Analysis of the feasibility of the idea:-
  - **Highly Feasible & Scalable:** By utilizing Scikit-Learn ensemble methods instead of heavy deep learning (LSTMs), the engine requires minimal compute overhead. It successfully runs real-time inference on entry-level edge hardware (e.g., Intel Core i3).
  - **Immediate Deployment:** The inclusion of a robust synthetic data generator allows security teams to test, validate, and tune the models immediately without waiting weeks for organic log collection.
- Potential challenges and risks:-
  - **Concept Drift:** Legitimate enterprise behavior changes over time (e.g., employees shifting departments, new devices), which could trigger false positives.
  - **The Cold-Start Problem:** Accurately scoring access events for brand-new employees or devices that possess no historical baseline.
- Strategies for overcoming these challenges:-
  - **Addressing Drift:** The architecture supports sliding-window retraining. The Isolation Forest can be cheaply retrained on a nightly batch of the last 7-14 days of logs to naturally adapt to evolving behavioral baselines.
  - **Addressing Cold-Starts:** The pipeline evaluates new events against the global enterprise baseline rather than strictly individual histories. Unseen categorical variables are handled safely via zero-indexed fallback encoders.

ARTIFACTS

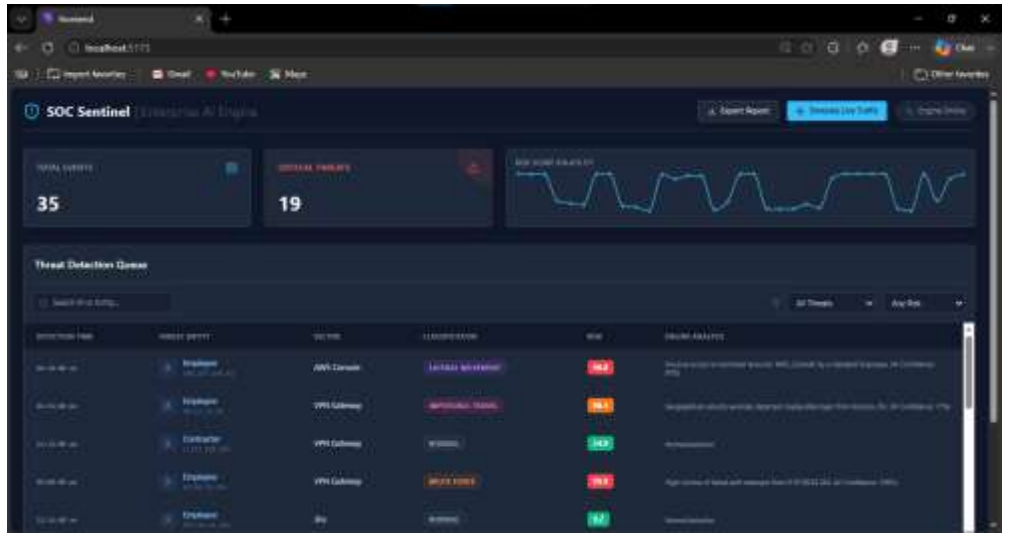
- detector.py(predict_proba code):-

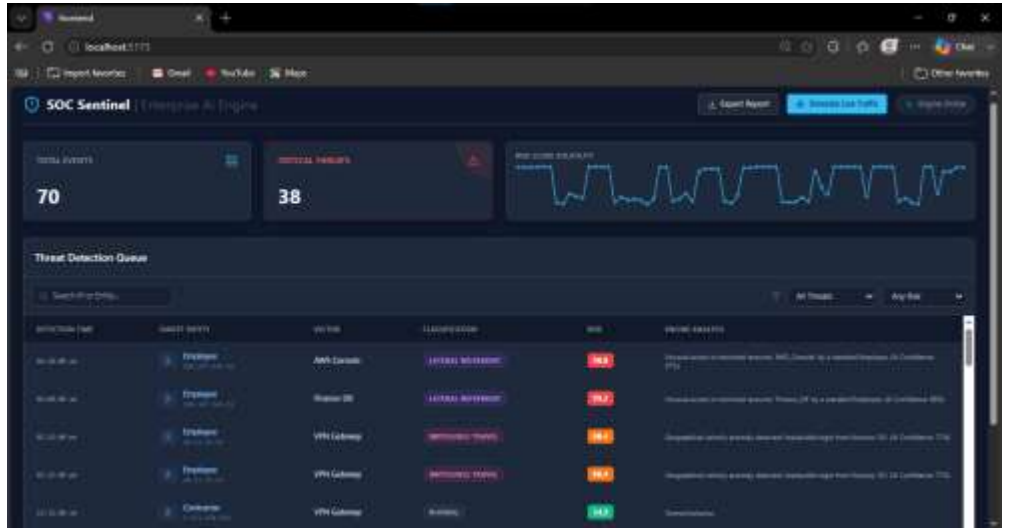


- Simulation Snap:-



- Dashboard Snap:-





RESEARCH AND REFERENCES

- **Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *Isolation Forest*. IEEE International Conference on Data Mining. Applied for baseline outlier detection.**
- **Scikit-Learn Developers (2024). *Ensemble Methods: Random Forests*. Used for supervised threat categorization and dynamic risk score mapping via model confidence (predict_proba).**
- **MITRE ATT&CK Framework. Used as the theoretical basis for defining the synthetic attack taxonomies (Lateral Movement, Brute Force, Impossible Travel).**
- **NIST Special Publication 800-207. Principles applied to continuous behavioral monitoring of authenticated sessions.**